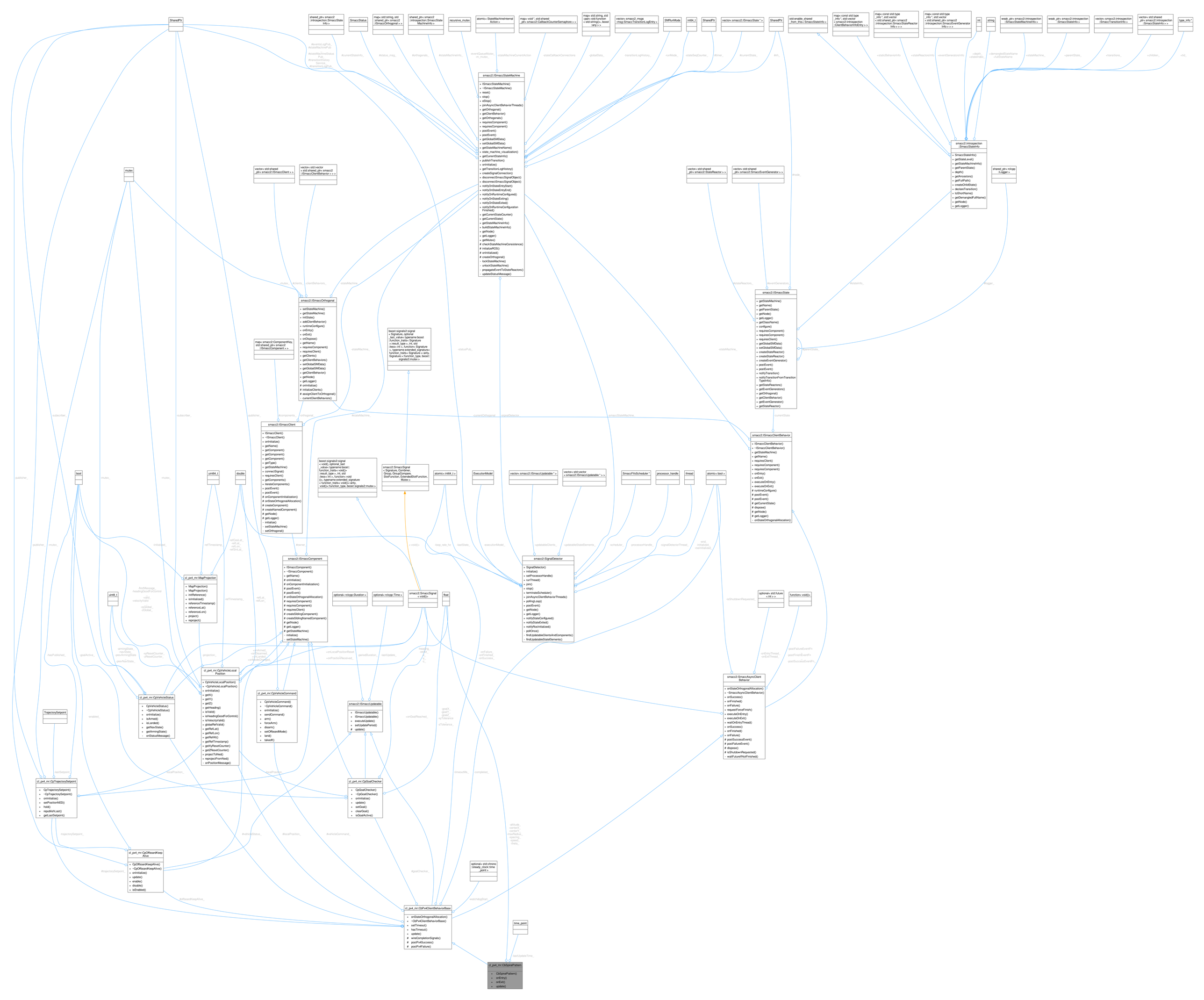
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2::msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
SharedPtr
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
string
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
#eventsLogPub_
#stateMachinePub_
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
eventQueueMutex_ m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#info_
#node_
+staticBehaviorInfo
+stateReactorsInfo
+eventGeneratorsInfo
+depth_ +stateInfoIdx_
+demangledStateName +fullStateName_
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# oninitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
shared_ptr< rclcpp ::Logger >
mutex
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
-mutex_
#clients_
-clientBehaviors_
-stateMachine_
#stateReactors_
#eventGenerators_
#stateInfo_
#logger_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# oninitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
-stateMachine_
-statusPub_
-stateMachine_
-parentState_
-components_
-orthogonal_
#stateMachine_
-currentOrthogonal_
-signalDetector_
-smaccStateMachine_
-currentState
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
+ onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
atomic< int64_t >
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
processor_handle
thread
atomic< bool >
bool
uint64_t
double
-publisher_
-mutex_
-subscriber_
-subscriber_
publisher_
-mutex_
-initialized_
-refTimestamp_
-refCosLat_ -refLat_ -refLon_ -refSinLat_
#owner_
< void()>
#loop_rate_hz_
-lastState_
-executionModel_
-updatableClients_
-updatableStateElements_
-scheduler_
-processorHandle_
-signalDetectorThread_
-end_ -initialized_ -isInitialized_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyPostInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
cl_px4_mrc::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
uint8_t
optional< rclcpp::Duration >
optional< rclcpp::Time >
smacc2::SmaccSignal < void()>
float
optional< std::future < int > >
function< void()>
-firstMessage_ -headingGoodForControl_
-valid_ -velocityValid_
-xyGlobal_ -zGlobal_
-refTimestamp_
-refLat_ -refLon_
#isShutdownRequested_
-onEntryThread_ -onExitThread_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn
-hasPublished_
-goalActive_
-armingState_ -navState_ -prevArmingState_
-prevNavState_
-xyResetCounter_ -zResetCounter_
projection_
+onArmed_ +onDisarmed_ +onLanded_ +onRotateOffboard_
+onLocalPositionReset_
+onPositionReceived_
-periodDuration_
-lastUpdate_
-headingSp_ -refX_ -refY_ -refZ_
-onFailure_ -onFinished_ -onSuccess_
cl_px4_mrc::ClpVehicleLocal Position
+ ClpVehicleLocalPosition()
+ ~ClpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlts()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
# onSuccess()
# onFinished()
# onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_px4_mrc::ClpVehicleStatus
+ ClpVehicleStatus()
+ ~ClpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
cl_px4_mrc::ClpVehicleCommand
+ ClpVehicleCommand()
+ ~ClpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
-goalX_ -goalY_ -goalZ_ -xyTolerance_
-zTolerance_
+onGoalReached_
TrajectorySetpoint
-enabled_
-lastSetpoint_
-localPosition_
-localPosition_
-timeoutMs_
-completed_
cl_px4_mrc::ClpTrajectorySetpoint
+ ClpTrajectorySetpoint()
+ ~ClpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
cl_px4_mrc::ClpGoalChecker
+ ClpGoalChecker()
+ ~ClpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
-trajectorySetpoint_
#vehicleStatus_
#localPosition_
#vehicleCommand_
altitude_ -centerX_ -centerY_ -maxRadius_ -spacing_ -speed_ -theta_
cl_px4_mrc::ClpOffboardKeep Alive
+ ClpOffboardKeepAlive()
+ ~ClpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
#trajectorySetpoint_
#offboardKeepAlive_
#goalChecker_
optional< std::chrono ::steady_clock::time _point >
-watchdogStart_
cl_px4_mrc::ClPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~ClPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
time_point
-lastUpdateTime_
cl_px4_mrc::ClpSpiralPattern
+ ClpSpiralPattern()
+ onEntry()
+ onExit()
+ update()